

Proof that the Set of Rational Numbers is not Order Complete

Statement

We shall prove that the ordered field $\mathbb{Q}$ of rational numbers is not order complete.

Recall the definition:

Definition (Order Completeness). An ordered set S is said to be order complete if every nonempty subset of S that is bounded above possesses a least upper bound (supremum) in S .

Thus, to show that $\mathbb{Q}$ is not order complete, it is enough to find:

- a nonempty subset $E \subset \mathbb{Q}$,
- such that E is bounded above in $\mathbb{Q}$,
- but E has no least upper bound in $\mathbb{Q}$.

Construction of the Set

Consider the set

$$E = \{x \in \mathbb{Q} : x^2 < 2\}.$$

We now analyze this set carefully.

Step 1: E is Nonempty

We observe that

$$1 \in E,$$

because

$$1^2 = 1 < 2.$$

Hence,

$$E \neq \emptyset.$$

Step 2: E is Bounded Above in $\mathbb{Q}$

We claim that 2 is an upper bound of E .

Indeed, let $x \in E$. Then

$$x^2 < 2.$$

Suppose $x \geq 2$. Then

$$x^2 \geq 4,$$

which contradicts $x^2 < 2$.

Therefore,

$$x < 2$$

for every $x \in E$.

Hence 2 is an upper bound of E in $\mathbb{Q}$.

Step 3: Assume E has a Supremum in $\mathbb{Q}$

Assume, for contradiction, that E possesses a least upper bound in $\mathbb{Q}$.

Let

$$\alpha = \sup E,$$

where $\alpha \in \mathbb{Q}$.

Since $1 \in E$ and α is an upper bound of E , we have

$$\alpha \geq 1.$$

Hence,

$$\alpha + 2 > 0.$$

We shall derive a contradiction.

Step 4: Analyze the Possible Values of α^2

There are three possibilities:

$$\alpha^2 < 2, \quad \alpha^2 > 2, \quad \alpha^2 = 2.$$

To analyze the first two cases, define

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2}.$$

Since $\alpha \in \mathbb{Q}$, it follows that $q \in \mathbb{Q}$.

Simplifying,

$$q = \frac{\alpha(\alpha + 2) - (\alpha^2 - 2)}{\alpha + 2} = \frac{2\alpha + 2}{\alpha + 2}.$$

Now compute:

$$q^2 - 2 = \left(\frac{2\alpha + 2}{\alpha + 2} \right)^2 - 2.$$

Therefore,

$$q^2 - 2 = \frac{4\alpha^2 + 8\alpha + 4 - 2(\alpha^2 + 4\alpha + 4)}{(\alpha + 2)^2}.$$

Hence,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2}.$$

Case 1: $\alpha^2 < 2$

Assume

$$\alpha^2 < 2.$$

Then

$$\alpha^2 - 2 < 0.$$

Since $\alpha + 2 > 0$,

$$\frac{\alpha^2 - 2}{\alpha + 2} < 0.$$

Hence,

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2} > \alpha.$$

Also,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2} < 0,$$

because $(\alpha + 2)^2 > 0$.

Thus,

$$q^2 < 2,$$

which implies

$$q \in E.$$

But $q > \alpha$, contradicting the fact that α is an upper bound of E .

Therefore,

$$\alpha^2 < 2$$

is impossible.

Case 2: $\alpha^2 > 2$

Assume

$$\alpha^2 > 2.$$

Then

$$\alpha^2 - 2 > 0.$$

Since $\alpha + 2 > 0$,

$$\frac{\alpha^2 - 2}{\alpha + 2} > 0.$$

Hence,

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2} < \alpha.$$

Also,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2} > 0.$$

Thus,

$$q^2 > 2.$$

Now let $x \in E$. Then

$$x^2 < 2 < q^2.$$

Since $q > 0$, it follows that

$$x < q.$$

Hence q is an upper bound of E .

But

$$q < \alpha,$$

contradicting the fact that α is the least upper bound of E .

Therefore,

$$\alpha^2 > 2$$

is impossible.

Case 3: $\alpha^2 = 2$

The only remaining possibility is

$$\alpha^2 = 2.$$

Assume

$$\alpha = \frac{m}{n},$$

where $m, n \in \mathbb{Z}$ have no common factor and $n \neq 0$.

Then

$$\left(\frac{m}{n}\right)^2 = 2,$$

so

$$m^2 = 2n^2.$$

Thus m^2 is even, which implies that m is even.

Let

$$m = 2k$$

for some integer k .

Substituting,

$$(2k)^2 = 2n^2,$$

which gives

$$4k^2 = 2n^2,$$

and hence

$$2k^2 = n^2.$$

Therefore n^2 is even, implying that n is even.

Thus both m and n are even, contradicting the assumption that $\frac{m}{n}$ is in lowest terms.

Hence no rational number satisfies

$$\alpha^2 = 2.$$

Therefore,

$$\alpha \notin \mathbb{Q},$$

contradicting our assumption that $\alpha \in \mathbb{Q}$.

Thus this case is also impossible.

Conclusion

All three possibilities lead to contradictions.

Hence our assumption that E possesses a supremum in $\mathbb{Q}$ is false.

Therefore, the set

$$E = \{x \in \mathbb{Q} : x^2 < 2\}$$

is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$.

Consequently,

$$\mathbb{Q}$$

is not order complete.